

FIG. 14.13
Example 14.1(a).

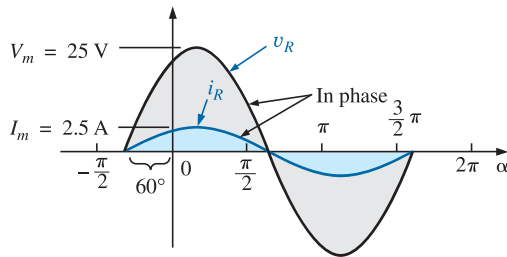


FIG. 14.14
Example 14.1(b).

to be considered. Simply applying Ohm's law, $I_m = E_m/X_L$ (or X_C), and keeping in mind the phase relationship between the voltage and current for each element, will be sufficient to complete the examples.

EXAMPLE 14.1 The voltage across a resistor is indicated. Find the sinusoidal expression for the current if the resistor is $10\ \Omega$. Sketch the curves for v and i .

- $v = 100 \sin 377t$
- $v = 25 \sin(377t + 60^\circ)$

Solutions:

$$\text{a. Eq. (14.2): } I_m = \frac{V_m}{R} = \frac{100\ \text{V}}{10\ \Omega} = 10\ \text{A}$$

(v and i are in phase), resulting in

$$i = 10 \sin 377t$$

The curves are sketched in Fig. 14.13.

$$\text{b. Eq. (14.2): } I_m = \frac{V_m}{R} = \frac{25\ \text{V}}{10\ \Omega} = 2.5\ \text{A}$$

(v and i are in phase), resulting in

$$i = 2.5 \sin(377t + 60^\circ)$$

The curves are sketched in Fig. 14.14.

EXAMPLE 14.2 The current through a $5\ \Omega$ resistor is given. Find the sinusoidal expression for the voltage across the resistor for $i = 40 \sin(377t + 30^\circ)$.

Solution: Eq. (14.3): $V_m = I_m R = (40\ \text{A})(5\ \Omega) = 200\ \text{V}$

(v and i are in phase), resulting in

$$v = 200 \sin(377t + 30^\circ)$$

EXAMPLE 14.3 The current through a $0.1\ \text{H}$ coil is provided. Find the sinusoidal expression for the voltage across the coil. Sketch the v and i curves.

- $i = 10 \sin 377t$
- $i = 7 \sin(377t - 70^\circ)$

Solutions:

$$\text{a. Eq. (14.4): } X_L = \omega L = (377\ \text{rad/s})(0.1\ \text{H}) = 37.7\ \Omega$$

$$\text{Eq. (14.5): } V_m = I_m X_L = (10\ \text{A})(37.7\ \Omega) = 377\ \text{V}$$

and we know that for a coil v leads i by 90° . Therefore,

$$v = 377 \sin(377t + 90^\circ)$$

The curves are sketched in Fig. 14.15.

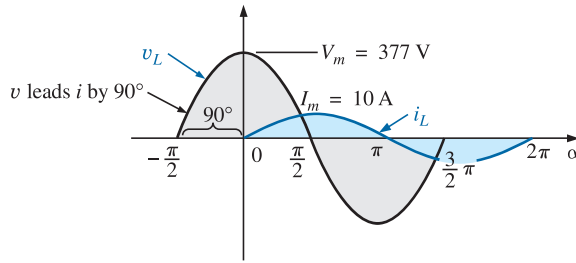


FIG. 14.15
Example 14.3(a).

b. X_L remains at 37.7Ω .

$$V_m = I_m X_L = (7 \text{ A})(37.7 \Omega) = 263.9 \text{ V}$$

and we know that for a coil v leads i by 90° . Therefore,

$$v = 263.9 \sin(377t - 70^\circ + 90^\circ)$$

and

$$v = 263.9 \sin(377t + 20^\circ)$$

The curves are sketched in Fig. 14.16.

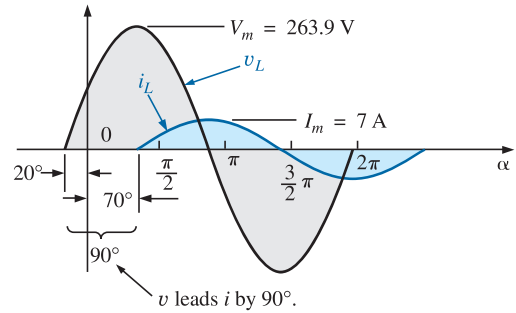


FIG. 14.16
Example 14.3(b).

EXAMPLE 14.4 The voltage across a 0.5 H coil is provided below. What is the sinusoidal expression for the current?

$$v = 100 \sin 20t$$

Solution:

$$X_L = \omega L = (20 \text{ rad/s})(0.5 \text{ H}) = 10 \Omega$$

$$I_m = \frac{V_m}{X_L} = \frac{100 \text{ V}}{10 \Omega} = 10 \text{ A}$$

and we know the i lags v by 90° . Therefore,

$$i = 10 \sin(20t - 90^\circ)$$

EXAMPLE 14.5 The voltage across a $1 \mu\text{F}$ capacitor is provided below. What is the sinusoidal expression for the current? Sketch the v and i curves.

$$v = 30 \sin 400t$$

Solution:

$$\text{Eq. (14.6): } X_C = \frac{1}{\omega C} = \frac{1}{(400 \text{ rad/s})(1 \times 10^{-6} \text{ F})} = \frac{10^6 \Omega}{400} = 2500 \Omega$$

$$\text{Eq. (14.7): } I_m = \frac{V_m}{X_C} = \frac{30 \text{ V}}{2500 \Omega} = 0.0120 \text{ A} = 12 \text{ mA}$$

and we know that for a capacitor i leads v by 90° . Therefore,

$$i = 12 \times 10^{-3} \sin(400t + 90^\circ)$$

The curves are sketched in Fig. 14.17.

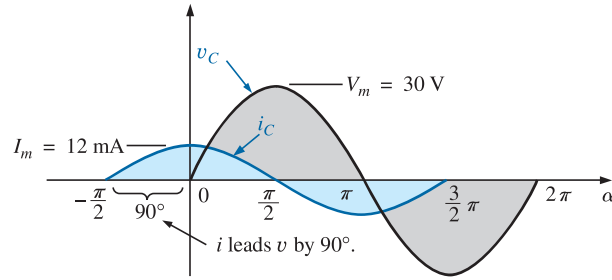


FIG. 14.17

Example 14.5.

EXAMPLE 14.6 The current through a $100\ \mu\text{F}$ capacitor is given. Find the sinusoidal expression for the voltage across the capacitor.

$$i = 40 \sin(500t + 60^\circ)$$

Solution:

$$X_C = \frac{1}{\omega C} = \frac{1}{(500\ \text{rad/s})(100 \times 10^{-6}\ \text{F})} = \frac{10^6\ \Omega}{5 \times 10^4} = \frac{10^2\ \Omega}{5} = 20\ \Omega$$

$$V_M = I_M X_C = (40\ \text{A})(20\ \Omega) = 800\ \text{V}$$

and we know that for a capacitor, v lags i by 90° . Therefore,

$$v = 800 \sin(500t + 60^\circ - 90^\circ)$$

and

$$v = \mathbf{800 \sin(500t - 30^\circ)}$$

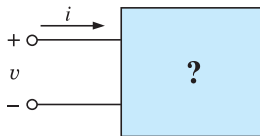


FIG. 14.18

Example 14.7.

EXAMPLE 14.7 For the following pairs of voltages and currents, determine whether the element involved is a capacitor, an inductor, or a resistor. Determine the value of C , L , or R if sufficient data are provided (Fig. 14.18):

- $v = 100 \sin(\omega t + 40^\circ)$
 $i = 20 \sin(\omega t + 40^\circ)$
- $v = 1000 \sin(377t + 10^\circ)$
 $i = 5 \sin(377t - 80^\circ)$
- $v = 500 \sin(157t + 30^\circ)$
 $i = 1 \sin(157t + 120^\circ)$
- $v = 50 \cos(\omega t + 20^\circ)$
 $i = 5 \sin(\omega t + 110^\circ)$

Solutions:

- a. Since v and i are *in phase*, the element is a *resistor*, and

$$R = \frac{V_m}{I_m} = \frac{100\ \text{V}}{20\ \text{A}} = \mathbf{5\ \Omega}$$

- b. Since v *leads* i by 90° , the element is an *inductor*, and

$$X_L = \frac{V_m}{I_m} = \frac{1000\ \text{V}}{5\ \text{A}} = 200\ \Omega$$

so that $X_L = \omega L = 200\ \Omega$ or

$$L = \frac{200 \, \Omega}{\omega} = \frac{200 \, \Omega}{377 \, \text{rad/s}} = \mathbf{0.53 \, \text{H}}$$

c. Since i leads v by 90° , the element is a *capacitor*, and

$$X_C = \frac{V_m}{I_m} = \frac{500 \, \text{V}}{1 \, \text{A}} = 500 \, \Omega$$

$$\text{so that } X_C = \frac{1}{\omega C} = 500 \, \Omega \text{ or}$$

$$C = \frac{1}{\omega 500 \, \Omega} = \frac{1}{(157 \, \text{rad/s})(500 \, \Omega)} = \mathbf{12.74 \, \mu\text{F}}$$

$$\begin{aligned} \text{d. } v &= 50 \cos(\omega t + 20^\circ) = 50 \sin(\omega t + 20^\circ + 90^\circ) \\ &= 50 \sin(\omega t + 110^\circ) \end{aligned}$$

Since v and i are *in phase*, the element is a *resistor*, and

$$R = \frac{V_m}{I_m} = \frac{50 \, \text{V}}{5 \, \text{A}} = \mathbf{10 \, \Omega}$$

14.4 FREQUENCY RESPONSE OF THE BASIC ELEMENTS

Thus far, each description has been for a set frequency, resulting in a fixed level of impedance for each of the basic elements. We must now investigate how a change in frequency affects the impedance level of the basic elements. It is an important consideration because most signals other than those provided by a power plant contain a variety of frequency levels. The last section made it quite clear that the reactance of an inductor or a capacitor is sensitive to the applied frequency. However, the question is, How will these reactance levels change if we steadily increase the frequency from a very low level to a much higher level?

Although we would like to think of every element as ideal, it is important to realize that every commercial element available today *will not respond in an ideal fashion for the full range of possible frequencies*. That is, each element is such that for a particular range of frequencies, it performs in an essentially ideal manner. However, there is always a range of frequencies in which the performance varies from the ideal. Fortunately, the designer is aware of these limitations and will take them into account in the design.

The discussion begins with a look at the response of the *ideal elements*—a response that will be assumed for the remaining chapters of this text and one that can be assumed for any initial investigation of a network. This discussion is followed by a look at the factors that cause an element to deviate from an ideal response as frequency levels become too low or high.

Ideal Response

Resistor R For an ideal resistor, you can assume that *frequency will have absolutely no effect on the impedance level*, as shown by the response in Fig. 14.19. Note that at 5 kHz or 20 kHz, the resistance of the resistor remains at $22 \, \Omega$; there is no change whatsoever. For the rest of

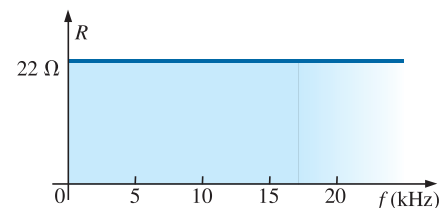


FIG. 14.19

R versus f for the range of interest.